

IMPACT-Scribe Human Study Protocol

This document defines only the three comparison conditions required by the current paper-facing study, so participants are not distracted by unrelated tool functionality.

Goal

Compare the efficiency and final annotation quality of the following three protocols under the same videos and the same starting conditions:

1. Manual annotation with another existing tool
2. IMPACT-Scribe `Scribble Only`
3. IMPACT-Scribe `Scribble + Planner`

Fairness Principles

- Use the same set of videos across all three conditions whenever possible.
- If an initial coarse segmentation or prelabels are provided, all three conditions must start from the same initialization.
- If the external manual tool cannot import prelabels, explicitly record that condition as `from scratch` and do not mix it with `with prelabels`.
- Each participant should complete a short practice session before the timed evaluation begins.

Condition A: Manual Annotation with Another Tool

This condition is not performed inside IMPACT-Scribe.

Recommended fields to record:

- start time and end time
- video ID
- whether initial prelabels were used
- exported annotation file path

Recommended reported metrics:

- `Boundary-F1@10 / @25 / @50`
- `edit score`
- `frame accuracy`

- completion time

Condition B: Scribble Only

Participant Actions

1. Open `app.py`.
2. Load the video, features, and existing annotation or prelabel.
3. Select `Study: Scribble Only` in the top `Study:` dropdown.
4. Draw boundary scribbles directly on the timeline.
5. If the proposal is unsatisfactory, add another scribble at the same location or drag the proposal boundary directly.
6. Click `Accept Boundary`.
7. Repeat until the current video is complete.

System Behavior in This Condition

- the study mode is fixed to `Coarse`
- `Fine`-related entry points are hidden
- the `Interaction` dropdown is hidden
- the planner is disabled
- the participant must decide where the next boundary should be

Condition C: Scribble + Planner

Participant Actions

1. Open `app.py`.
2. Load the video, features, and existing annotation or prelabel.
3. Select `Study: Scribble + Planner` in the top `Study:` dropdown.
4. Click `Next Boundary`.
5. The system jumps to the next suggested boundary.
6. Draw a boundary scribble on the timeline; if needed, add another scribble or drag the proposal boundary.
7. Click `Accept Boundary`.
8. Click `Next Boundary` again, or continue according to the current interface flow to the next suggested boundary.
9. Repeat until the current video is complete.

System Behavior in This Condition

- the study mode is fixed to `Coarse`
- `Fine`-related entry points are hidden
- the `Interaction` dropdown is hidden
- the planner recommends only `boundary` queries and does not mix in label review or state repair

Boundary Semantics in the Current Study Mode

The current paper-facing study uses a single unified `boundary scribble` gesture:

- scribbling in blank space means creating a new boundary
- a short and narrow scribble centered on an existing boundary means deleting that boundary and merging the adjacent segments
- a wider scribble crossing an existing boundary means refining that boundary

All of these cases reuse the same `Accept Boundary` flow.

Blank-Canvas Semantics

On a blank canvas, the system works as a continuous temporal partition rather than a set of isolated patches:

- the first accepted boundary fills the immediately preceding unlabeled span
- the second accepted boundary fills the next adjacent unlabeled span
- the already visited part of the timeline should not be left with gaps

In other words, blank-canvas interaction should behave like progressive forward segmentation and filling along the timeline.

Selection and Drawing Behavior

To reduce accidental edits, the current interaction priority is:

- clicking an existing segment or marker selects it first
- a scribble only starts after the drag distance passes the threshold
- right click preserves the delete semantics for existing annotations or markers

Therefore, participants should not sketch many locations first and accept them later in a batch. They should process one boundary at a time:

1. draw or adjust the current boundary

2. inspect the proposal
3. click `Accept Boundary`
4. move on to the next boundary

Label Assistance

The current paper-facing study remains boundary-first, but label assistance is already integrated into the workflow:

- the left panel shows recommended labels for the current segment or gap
- for `blank fill` and `remove boundary` proposals, the recommended label can be overridden before clicking `Accept Boundary`
- candidate labels come from prototype memory, imported candidates, optional text priors, and runtime confusion memory

Notes:

- in study mode, the left label panel title is shown as `Recommended Labels / Labels`
- this label path is only an aid and is not the primary workflow participants must learn first

Main IMPACT-Scribe Entry Points Used in the Study

Only the following in-tool entry points should be used during the human study:

- `Study:` dropdown
for switching between `Standard / Scribble Only / Scribble + Planner`
- `Next Boundary`
used only in `Scribble + Planner` to jump to the next suggested boundary
- `Boundary Scribble` on the timeline
- `Accept Boundary` in the footer

Recommended Logging Fields

For each session, record:

- `participant_id`
- `condition`
- `video_id`
- `start_time`
- `end_time`

- `total_elapsed_seconds`
- `interaction_count`
- `accept_count`
- `reject_count`
- `second_correction_count`
- `output_annotation_path`
- `output_scribble_sidecar_path`

IMPACT-Scribe records `study_condition` in the correction session and scribble sidecar so later analysis can group results by condition.

Recommended Comparisons

Two primary comparisons are recommended:

1. `manual annotation with another tool` vs `Scribble Only`
to evaluate whether boundary-scribble interaction itself reduces effort.
2. `Scribble Only` vs `Scribble + Planner`
to evaluate whether the planner reduces search cost and improves return per unit time.

Do not use `manual annotation with another tool` and `Scribble + Planner` as the only direct comparison. `Scribble Only` must remain in the middle, otherwise the planner contribution cannot be isolated.